

1 Legend for later use:

In this section we will begin by outlining some of the common notation with further notation introduced in the literature review. ...

Suppose that we have a statistical model $P_\theta \in \mathcal{P}$ with the parameters $\theta \in \Theta$ for the data points $x_{1:n} = \{x_1, x_2, \dots, x_n\}$ with theoretical and often unknown data generating process (DGP) $D(x)$. These parameters typically(?) represent information about the data as we follow statistical methods to model the data. In frequentist statistics these parameters θ are treated as constants ~~post-learning~~ (need a better term than learning), however, when dealing with the bayesian approach, the parameters are treated as random variables / distributions. In the most standard bayes we have the posterior density

$$\pi_B(\theta|x_{1:n}) = \frac{\pi(\theta)L(x_{1:n}|\theta)}{m(x_{1:n})}.$$

Where $\pi(\theta)$ is our prior knowledge of the parameters θ and $\pi_B(\theta|x_{1:n})$ is the pdf of the distribution $\theta|x_{1:n}$. The B here indicating that this is the standard bayesian posterior. Here $m(x_{1:n}) = \int_{\Theta} L(x_{1:n}|\theta)d\pi(\theta)$ is the marginal normalising constant of the distribution, whose computation often makes the distribution intractable.

Bayesian inference ~~using the this method~~ is conceptually and practically appealing as, unlike frequentist statistic methods, we are able to allow our previous knowledge, specific domain expertise and experts in the subject matter to further inform our distribution. This injection of previous knowledge comes in the form of a carefully specified prior $\pi(\theta)$. Naturally, this comes with the potential risk of poor prior choice.

...

Further, bayesian posteriors producing belief distributions (as opposed to point estimates, as an example, the direct outputs of MLE) over the parameters of interest $\theta \in \Theta$. This has the advantage that, during inference where we integrate across the posterior $\pi_B(\theta|x_{1:n})$... THIS NEEDS CONTINUING ABOUT THE BENEFIT OF THE POSTERIOR HAVING belief distributions over normal point estimates. namely that we are able to "quantify Uncertainty"(this is not true / im not too sure we need a paper for this, however, we do know that it enables) the distributions have the ability to not be over-confident in our *best*(make a [1] here noting that *best* here refers to the best of other point estimates), as rather than using a single value of $\theta \in \Theta$ we use can make a prediction through a conventional Bayesian approach to prediction uses the posterior distribution to integrate out parameters in a density for unobserved data conditional on the observed data:

1. integrate over the model P_{θ} to get an output of the model.
2. Integrate over a target with standard items of the distribution such as $\hat{\theta} = \int \theta d\pi_B = \int \theta \pi_B(\theta|x_{1:n})d\theta$

In typical bayes the $L(x|\theta)$ is the likelyhood of the data and parameters. bruh

Moving to a more generalised bayes and we can change this to an optimisation centric approach. generalised bayes moves the likelihood to a into a loss function that is raised in an exponential.

$$\begin{aligned}\pi_w(\theta | x_{1:n}) &= \frac{\exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta)}{\int_{\Theta} \exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta) d\theta} \\ &\propto \exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta)\end{aligned}$$

where once again there is a normaliser in the denominator that without we can find the distribution up to proportionality. Here also exists a $w \in \mathbb{R}$ that represents a learning rate/loss score. This posterior is the Bissiri-Holmes-Walker (2016) Gibbs posterior.

It is important to note here that from this posterior we can recover our previous standard bayes posterior π_B in the specific case with $\ell(\theta, x_i) = -\log p(x_i|\theta)$ and the learning rate as $w = 1$. Following this we get

$$\begin{aligned}\pi_w(\theta | x_{1:n}) &\propto \exp\left\{-w \sum_{i=1}^n \ell(\theta, x_i)\right\} \pi(\theta) \\ &= \exp\left\{\sum_{i=1}^n \log p(x_i | \theta)\right\} \pi(\theta) \\ &= \left(\prod_{i=1}^n p(x_i | \theta)\right) \pi(\theta) \\ &= L(x_{1:n} | \theta) \pi(\theta)\end{aligned}$$

And further the normalising constant recovers the marginal likelihood in the same manner. Hence we recover $\pi_B(\theta|x_{1:n})$.

Aside on the added learning rate term: Looking at the learning rate w here, we can see this as as weighting between the prior and the data and a method against model misspecification. assuming that the prior is well specified. It acts as a effective-sample-size multiplier effectively updating the the prior with only wN data points The effect of this learning rate is exact and analytic and *does* effect the final distribution of the posterior, whilst depending on the algorithm can have an effect on the algorithm that approximates it. *note that is if the algorithm is not an analytical solver, it is also different from the algorithmic step size in an optimiser. *Later with actual step sizes in optimisers, we will not have this clean split between the data weight and the optimisers effect on the distribution.

Further notes on the loss function, coherence and moving to an optimisation statement: Bissiri, Holmes & Walker have a heavy emphasis on coherence which states that

... update rule for coherence... *etc on importance. Not all loss functions hold to this as while the standard log loss does, as we build up to other ones this may not hold. The paper breaks down the. Often, researchers are only interested in a simple, low-dimensional statistic, such as a population mean or median. In standard Bayesian frameworks, estimating these simple statistics frustratingly still requires modeling the entire data distribution. Furthermore, some parameters of interest do not directly index any standard parametric family of density functions. (ADD IN STATEMENT ABOUT HOW IN OUR CASE FOR THE PAPER WE ARE interested in the full posterior as it is still good and need). An example is using an absolute error loss function to robustly estimate a population median without specifying a distribution. (Can this still surmount model misspecification?) (Can we also do an eventual link here to Posterior oriented prediction, would the analytical result be the same if we aimed for something similar?)

Here there is also a pivot ... ADD IN HOW THIS STARTS TO PIVOT TO OPTIMISATION CENTRIC. view .

Knoblauch et al. generalise this into a triple of loss ℓ divergence and a feasible set Π of distributions to optimise over ...

There are a few requirements this must have

(A general framework for updating belief distributions (Bissiri et al.))

2 Lit review

1. Foundations of Belief and Updating & Variational Inference

- A general framework for updating belief distributions (Bissiri et al.)
- Stochastic Variational Inference
- An optimisation-centric view on Bayes' rule: Reviewing and generalising variational inference (Knoblauch et al.)

2. The Shift to Predictively Oriented Posteriors

- Predictive Variational Inference: Learning the predictively optimal posterior distribution
- Predictive Variational Inference for flexible regression models
- Predictively Oriented Posteriors (McLachlan et al.)

3. Particle-Based Inference, MMD, and Misspecification

- Stein Variational Gradient Descent (Liu & Wang) (Very quick mention) and time complexity things

- Prediction-Centric Uncertainty Quantification via MMD (Shen et al.)
- maybe: (Detecting model misspecification in Bayesian inverse problems via variational gradient descent)

4. **Variational Inference with Normalising Flows?? Adaptive Heterogeneous Mixtures of Normalising Flows for Robust Variational Inference?? And more MOE papers...**

(A general framework for updating belief distributions, Bissiri et al.)